Recovery Console Functional Requirements

Software Requirements Specification

IEEE 830-1998 outline

DRAFT — human review required

Version 1 | 2026-11-08T08:22:19.597980+00:00

Contents

1 Introduction 4
1.1 Purpose . . . 4
1.2 Scope . . . 4
1.3 Definitions, acronyms and abbreviations . . . 4
1.4 References . . . 4
1.5 Overview . . . 4
2 Overall description 4
2.1 Product perspective . . . 4
2.2 Product functions . . . 4
2.3 User characteristics and stakeholders . . . 5
2.4 Constraints . . . 5
2.5 Assumptions and dependencies . . . 5
3 Specific requirements 5
3.1 External interface requirements . . . 5
3.1.1 User interfaces . . . 5
3.1.2 Hardware interfaces . . . 5
3.1.3 Software interfaces . . . 5
3.1.4 Communications interfaces . . . 5
3.2 Functional requirements . . . 6
3.3 Performance requirements . . . 6
3.4 Logical database requirements . . . 6
3.5 Design constraints . . . 6
3.6 Software system attributes . . . 6
3.6.1 Reliability . . . 6
3.6.2 Availability . . . 6
3.6.3 Security . . . 6
3.6.4 Maintainability . . . 6
3.6.5 Portability . . . 6
3.7 Other requirements . . . 6
A Derived threat hypotheses 7
A.1 THR-001: Source-required records . . . 7
A.2 THR-002: Security and administrative audit evidence . . . 7
B Risk assessment 7
B.1 RISK-001 . . . 7
B.2 RISK-002 . . . 8
B.3 RISK-003 . . . 8
C SDLC recommendation and delivery workflow 8
C.1 Ranking and source evidence . . . 9
C.1.1 DevSecOps suitability reasons . . . 9
C.1.2 Incremental suitability reasons . . . 9
C.1.3 Hybrid suitability reasons . . . 9
C.1.4 Agile suitability reasons . . . 9
C.1.5 Spiral suitability reasons . . . 9
C.1.6 V-Model suitability reasons . . . 9
C.1.7 Waterfall suitability reasons . . . 10
C.2 Methodology phases . . . 10
C.2.1 Requirements and verification planning . . . 10
C.2.2 Design and test design . . . 10
C.2.3 DevSecOps delivery cycle . . . 11

C.2.4 Release and handoff 11

1 Introduction

1.1 Purpose

This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

This SRS describes the software's required behavior for users, administrators and the Project Analyst reviewing the specification.

1.2 Scope

Design a recovery console that lets archive operators initiate a restore of a saved archive. Archive operators can view the status of active restore jobs. The system must record restore requests in an audit log. Leave the audit log retention period open for review.

1.3 Definitions, acronyms and abbreviations

Term	Meaning
Audit record	A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.
Log	A collection of event records. This term does not itself require additional logging fields, events or access permissions.
Access restoration	Returning access that was previously restricted, within the source-defined permission boundary.
Retention	The period or rule for keeping records. A bare duration does not itself specify deletion timing or minimum/maximum bounds.

1.4 References

Reference	Source
Format	IEEE Std 830-1998, clauses 5 and Annex A (document organization).

1.5 Overview

Section 2 summarizes the product, stakeholders and open decisions. Section 3 contains the uniquely identified requirements. Detailed evidence, acceptance procedures are available in the Normal report. Appendices A-C contain derived threat hypotheses, risk assessment and the proposed SDLC workflow.

2 Overall description

2.1 Product perspective

Design a recovery console that lets archive operators initiate a restore of a saved archive.

2.2 Product functions

Functional topics are summarized below; section 3 retains the full requirements.

- allow archive operators to initiate a restore of a saved archive - FR-001

- allow archive operators to view the status of active restore jobs - FR-002
- record restore requests in an audit log - FR-003

2.3 User characteristics and stakeholders

Labels group stakeholders for readability. They do not combine permissions.

Education, training and experience requirements remain unspecified.

Role	Source user groups	Responsibility
Operator	operators	Performs only the role's source-defined operations in section 3.
Project Analyst	Document review role	Reviews requirements and tracks open decisions; this role grants no product access.

2.4 Constraints

No additional implementation constraints were confirmed. Source-defined restrictions remain in the requirements in section 3.

2.5 Assumptions and dependencies

Open decisions requiring confirmation:

- Leave the audit log retention period open for review.
- Which fields must the audit logs contain?
- What retention period is required for the audit logs?

3 Specific requirements

Each requirement is stated once with its original identifier and wording. Cross-references identify requirements covering more than one concern.

3.1 External interface requirements

3.1.1 User interfaces

No specific interface requirement was supplied; confirmation is required.

3.1.2 Hardware interfaces

No specific interface requirement was supplied; confirmation is required.

3.1.3 Software interfaces

No specific interface requirement was supplied; confirmation is required.

3.1.4 Communications interfaces

No specific interface requirement was supplied; confirmation is required.

3.2 Functional requirements

FR-001 · Must · explicit

The system shall allow archive operators to initiate a restore of a saved archive.

Traceability: E-001, TEST-001

FR-002 · Must · explicit

The system shall allow archive operators to view the status of active restore jobs.

Traceability: E-002, TEST-002

FR-003 · Must · explicit

The system shall record restore requests in an audit log.

Traceability: E-003, TEST-003

3.3 Performance requirements

No separate requirement was specified in the supplied brief.

3.4 Logical database requirements

No separate requirement was specified in the supplied brief.

3.5 Design constraints

No separate requirement was specified in the supplied brief.

3.6 Software system attributes

3.6.1 Reliability

No separate requirement was specified in the supplied brief.

3.6.2 Availability

No separate requirement was specified in the supplied brief.

3.6.3 Security

Also covered by: FR-003.

3.6.4 Maintainability

No separate requirement was specified in the supplied brief.

3.6.5 Portability

No separate requirement was specified in the supplied brief.

3.7 Other requirements

No separate requirement was specified in the supplied brief.

A Derived threat hypotheses

These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

These are planning hypotheses, not observed incidents or additional approved controls.

A.1 THR-001: Source-required records

Scenario: Recording or retention fails for the records required by the source, leaving the expected evidence unavailable. Source duty: The system must record restore requests in an audit log.

Hypothetical actor or cause: Unconfirmed actor or cause (planning hypothesis)

Potential consequence: Missing records could prevent review of the events covered by the stated record duty.

Linked requirements: FR-003

Original evidence: E-003

Residual exposure: Linked obligations define intended behavior, not proven prevention. Enforcement, misuse of permitted access and operational assumptions remain unverified until design and acceptance evidence are reviewed.

A.2 THR-002: Security and administrative audit evidence

Scenario: Derived hypothesis: An attacker or recording failure suppresses or alters source-required audit evidence so that access decisions, security events or admin actions cannot be reconstructed.

Hypothetical actor or cause: Hypothetical attacker, privileged insider or failure cause

Potential consequence: Investigators could lose reliable evidence of the events required by the brief.

Linked requirements: FR-003

Original evidence: E-003

Residual exposure: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

B Risk assessment

Risk treatments are proposals for review. Unknown likelihood and impact ratings mean unassessed; no numeric confidence or severity is inferred.

- Decision Q-07e31252fd21 is TBD.

B.1 RISK-001

Derived risk affecting Source-required records: Missing records could prevent review of the events covered by the stated record duty.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Recording or retention fails for the records required by the source, leaving the expected evidence unavailable. Source duty: The system must record restore requests in an audit log.

Proposed treatment: Review the failure or abuse path against the linked source obligations. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria

in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-003

Original evidence: E-003

B.2 RISK-002

Derived risk affecting Security and administrative audit evidence: Investigators could lose reliable evidence of the events required by the brief.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: An attacker or recording failure suppresses or alters source-required audit evidence so that access decisions, security events or admin actions cannot be reconstructed.

Proposed treatment: Compare source-required events with the resulting audit evidence; assess missing and altered-record cases and confirm unprovided protection or retention decisions. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Linked requirements: FR-003

Original evidence: E-003

B.3 RISK-003

An unresolved source decision may delay validation: Leave the audit log retention period open for review.

Likelihood: unknown; impact: unknown; owner: TBD.

Rationale: The source leaves this decision unresolved; no delivery or technical conclusion is inferred.

Proposed treatment: Obtain a stakeholder decision before baselining affected behavior; retain the unknown and revalidate resulting changes.

Linked requirements:

Original evidence: E-004

C SDLC recommendation and delivery workflow

Proposed lifecycle: DevSecOps

The DevSecOps model is selected based on derived operational validation evidence (E-003, E-004) indicating that monitoring and alert behavior must be validated with real-world feedback. This factor provides a distinct advantage over Incremental, Hybrid, and Agile models by directly linking continuous security integration to observable system performance. The score of 48 outperforms all other candidates due to the explicit derivation of operational validation as a differentiating input. However, no evidence addresses budget, delivery, failure consequences, feedback loops, legacy systems, regulation, schedule, stability, uncertainty, or verification—these are explicitly acknowledged as unknown factors and do not influence this decision. The recommendation remains provisional, requiring confirmation of team capacity and delivery constraints before adoption. No other model demonstrates a superior alignment with the derived operational validation basis.

Recommendation requires human confirmation of delivery constraints and team capacity. Suitability points are deterministic rule scores, not probabilities.

Policy version: sequential-sdlc-2.0

Unconfirmed decision factors: budget, delivery, failure consequences, feedback, legacy, regulation, schedule, stability, uncertainty, verification

- Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.
- Proposed suitability recommendation; confirm delivery constraints and team capacity before adopting this lifecycle. Missing factors are not assumed.

C.1 Ranking and source evidence

- operational_validation = monitoring feedback (derived). Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. Original evidence: E-003, E-004.

Candidate	Suitability points
DevSecOps	48
Incremental	46
Hybrid	45
Agile	40
Spiral	40
V-Model	40
Waterfall	40

C.1.1 DevSecOps suitability reasons

- Derived operational_validation (+8 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003, E-004)

C.1.2 Incremental suitability reasons

- Derived operational_validation (+6 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003, E-004)

C.1.3 Hybrid suitability reasons

- Derived operational_validation (+5 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003, E-004)

C.1.4 Agile suitability reasons

- No differentiating input evidence

C.1.5 Spiral suitability reasons

- No differentiating input evidence

C.1.6 V-Model suitability reasons

- No differentiating input evidence

C.1.7 Waterfall suitability reasons

- No differentiating input evidence

C.2 Methodology phases

Activities reference the canonical requirements and proposed verification work. Human gates are pending decisions, not approval records.

C.2.1 Requirements and verification planning

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review and baseline the source-defined work group: FR-001: The system shall allow archive operators to initiate a restore of a saved archive.; FR-002: The system shall allow archive operators to view the status of active restore jobs.; FR-003: The system shall record restore requests in an audit log.
- Activity: Record unresolved source decision: Leave the audit log retention period open for review.
- Activity: Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.
- Deliverable: Versioned requirements baseline
- Deliverable: Unit, integration, system and acceptance test plan
- Deliverable: Unresolved-decision record
- Entry criterion: Authoritative inputs and canonical requirements are available for review.
- Exit criterion: A human reviewer has recorded the baseline decision and disposition of unresolved scope.
- Security check: Review design and test evidence for violations of source-defined permissions and exclusions.

C.2.2 Design and test design

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Design the implementation for the source-defined work group: FR-001: The system shall allow archive operators to initiate a restore of a saved archive.; FR-002: The system shall allow archive operators to view the status of active restore jobs.; FR-003: The system shall record restore requests in an audit log.
- Activity: Refine the early unit, integration, system and acceptance test plan without choosing unresolved product technologies.
- Deliverable: Proposed design linked to canonical requirements
- Deliverable: Unit, integration, system and acceptance test plan
- Entry criterion: The requirements baseline and early test plan have been reviewed.
- Exit criterion: A human reviewer has recorded the design decision and test-plan readiness.

- Security check: Review design and test evidence for violations of source-defined permissions and exclusions.

C.2.3 DevSecOps delivery cycle

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review and prioritize the backlog for the next iteration without changing source-defined scope.
- Activity: Implement the source-defined work group: FR-001: The system shall allow archive operators to initiate a restore of a saved archive.; FR-002: The system shall allow archive operators to view the status of active restore jobs.; FR-003: The system shall record restore requests in an audit log.
- Activity: In an authorized isolated test environment, execute TEST-001: Given: An archive operator with valid credentials and access rights When: The operator selects a saved archive and initiates a restore request Expected: A restore job is created and the system acknowledges the request
- Activity: In an authorized isolated test environment, execute TEST-002: Given: An archive operator with valid credentials and access rights When: The operator requests to view active restore job statuses Expected: The operator receives a list of active restore jobs with their current status
- Activity: In an authorized isolated test environment, execute TEST-003: Given: An archive operator initiates a restore request When: The system processes the restore request Expected: The system records the restore request in an audit log
- Activity: Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
- Activity: Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.
- Activity: Review security-check evidence for every delivery change and record a human release decision; keep unresolved tooling and operational choices open.
- Deliverable: Requirement-linked implementation increment
- Deliverable: Recorded test evidence
- Deliverable: Reviewed feedback and next-iteration backlog
- Entry criterion: Prior work, source scope and the reviewed test plan are available for iteration planning.
- Exit criterion: A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.
- Security check: Review design and test evidence for violations of source-defined permissions and exclusions.

C.2.4 Release and handoff

Proposed roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Linked requirements: FR-001, FR-002, FR-003

Human gate: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

- Activity: Review the exact requirements baseline, implementation links and verification evidence before a release decision.
- Activity: Carry unresolved source decisions into handoff; generated text is not human approval.
- Deliverable: Requirements and verification evidence package
- Deliverable: Pending human release decision
- Deliverable: Outstanding-decision record
- Entry criterion: Verification evidence and unresolved decisions are available for human review.
- Exit criterion: A human has recorded the release/handoff decision for the reviewed content.
- Security check: Review design and test evidence for violations of source-defined permissions and exclusions.

Detailed acceptance criteria, evidence, architecture and the SDLC plan are available in the supporting report and canonical JSON.

Canonical content SHA-256: f11666f67ba776f4df1de37da4fa26f8af52bda25753d0821ba1667af664a74e